

DE KALB COUNTY AP, IN, USA

WMO: 720543

Lat: 41.300N

Lon: 85.067W

Elev: 880

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00167

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	6.6	-8.8	3.5	4.7	-3.9	4.6	8.9	22.8	27.7	20.2	25.3	4.1	250	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	90.3	71.7	88.0	71.2	85.7	69.5	75.6	85.5	74.0	83.3	72.3	80.9	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	124.2	81.1	71.7	121.0	80.0	69.9	113.8	78.1	39.6	86.1	38.2	83.1	36.5	80.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	17.2	15.3	-6.0	94.5	7.1	5.0	-11.1	98.1	-15.3	101.0	-19.3	103.8	-24.4	107.5
			-6.7	77.5	7.2	2.0	-11.9	79.0	-16.1	80.2	-20.2	81.3	-25.4	82.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	25.1	28.1	38.0	48.9	61.6	69.8	72.9	70.7	64.8	52.4	40.3	31.5
	DBStd	18.87	11.23	10.95	10.29	8.72	8.70	6.04	5.96	5.16	7.36	8.83	9.13	9.36
	HDD50	2885	772	617	395	122	11	0	0	0	1	77	314	576
	HDD65	6079	1236	1033	837	488	171	24	6	12	93	400	742	1037
	CDD50	3054	1	4	24	89	369	593	711	640	444	153	22	4
	CDD65	772	0	0	1	4	64	167	252	187	86	11	0	0
	CDH74	6978	0	0	12	56	661	1412	2393	1584	751	108	1	0
	CDH80	2181	0	0	0	2	185	433	876	435	235	15	0	0
Wind	WSAvg	5.9	7.4	7.3	7.2	7.9	5.8	4.9	3.9	3.5	4.2	5.5	6.2	6.6
Precipitation	PrecAvg	38.0	2.5	1.9	2.4	3.5	4.2	3.7	4.2	4.2	3.1	3.0	3.0	2.5
	PrecMax	50.3	4.8	5.6	4.7	6.1	7.4	6.0	7.2	10.8	6.9	8.7	6.5	5.9
	PrecMin	29.8	0.9	0.4	0.7	0.3	0.7	1.5	1.6	1.7	0.4	0.2	0.7	0.8
	PrecStd	4.8	1.2	1.2	0.9	1.5	2.0	1.4	1.5	2.1	1.9	2.0	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	63.3	77.1	79.5	89.6	91.4	97.1	90.5	90.9	81.8	70.4	61.1
		MCWB	53.4	51.1	62.6	59.3	68.9	71.1	74.2	71.2	71.3	66.1	56.6	56.5
	2%	DB	50.4	55.0	65.6	74.6	85.6	88.3	90.6	88.0	87.5	77.1	63.8	54.9
		MCWB	48.2	50.4	55.8	58.0	66.0	70.5	72.9	71.0	71.4	63.9	54.3	51.5
	5%	DB	45.1	48.4	59.5	70.1	81.4	85.7	88.1	84.4	81.6	72.2	59.4	49.8
		MCWB	42.2	44.4	50.9	55.7	65.2	69.4	72.1	69.2	68.5	61.1	50.9	46.7
	10%	DB	39.2	42.7	54.7	64.1	77.3	81.6	85.6	81.7	78.6	68.0	54.4	44.6
		MCWB	36.2	38.0	47.9	52.6	63.2	66.9	70.3	68.1	65.3	59.2	47.9	41.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	54.7	62.8	63.5	72.7	76.7	78.4	75.8	75.9	71.2	60.6	57.5
		MCDB	56.9	58.6	75.8	73.3	83.3	87.6	88.9	85.4	86.0	78.4	67.6	61.2
	2%	WB	48.0	50.2	57.9	60.2	69.6	73.6	76.4	74.0	73.1	66.4	56.0	52.6
		MCDB	50.0	55.4	64.2	68.8	79.5	84.1	86.5	82.5	82.2	72.6	62.4	54.3
	5%	WB	42.2	44.8	53.7	57.7	67.3	71.7	74.6	72.3	71.2	63.2	52.4	46.1
		MCDB	45.1	48.3	57.4	67.0	77.5	81.1	84.0	80.1	79.2	68.3	58.5	48.2
	10%	WB	37.0	38.7	48.7	55.0	65.6	69.8	72.8	70.9	68.8	60.2	48.1	41.3
		MCDB	39.4	42.2	55.9	64.6	74.5	77.2	81.5	78.2	74.9	67.4	54.7	44.3
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.1	16.8	20.3	20.8	19.7	20.1	20.5	20.6	19.0	15.2	11.8
		MCDBR	17.5	21.5	23.3	27.4	25.6	23.6	22.4	23.4	23.7	24.8	23.0	18.3
		MCWBR	15.3	16.6	16.1	15.5	12.2	9.9	8.4	9.5	10.6	13.3	15.1	15.5
	5% WB	MCDBR	16.8	21.2	20.3	23.1	21.0	19.4	19.1	17.6	19.7	20.5	20.9	17.5
		MCWBR	15.6	17.1	15.4	16.2	11.7	10.4	9.0	8.9	9.8	13.5	15.3	16.0
Clear-Sky Solar Irradiance	taub	0.300	0.333	0.350	0.400	0.437	0.441	0.437	0.441	0.400	0.352	0.320	0.302	
	taud	2.393	2.290	2.343	2.217	2.160	2.191	2.218	2.193	2.287	2.416	2.456	2.426	
	Ebn at Noon	270	275	285	276	267	266	266	260	264	266	261	259	
	Edh at Noon	26	33	36	43	47	46	44	44	37	29	24	23	
All-Sky Solar Radiation	RadAvg	490	771	1114	1469	1698	1907	1927	1692	1384	918	585	412	
	RadStd	50	103	100	132	157	128	104	84	115	85	81	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page